

some headings or markings on to the screen.

2. **Button:** This will let you trigger the event and is a very essential component.
3. **Horizontal arrangement:** This comprises special components that keep all child components aligned within themselves.
4. **Table arrangement:** This comprises special components that keep all child components, within itself, well arranged in rows and columns as specified in the properties.
5. **Clock:** This is the timer component that provides the time control of the events happening. Using this, we can set certain events to happen after a particular interval.
6. **Notifier:** This is used to display some instructions or give controls over your existing components. You will see its functionality in more detail as we implement it in our game.
7. **Canvas:** This is the component for drawing and animation. You can place various animation objects on the canvas to control them via user actions. You will be able to see more details on this aspect as we work on the application.
8. **Image:** The image component is used when you need to show some pictures or special effects on screen.
9. **TinyDB:** This is the database component for your application.

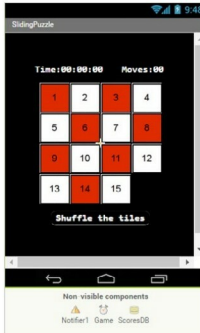


Figure 1: Designer screen

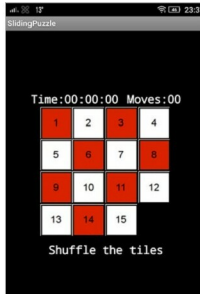


Figure 2: How the application looks

If we need to store some variables and values that we want to persist even after the application is closed, we use TinyDB.

The components that we need for this application are given in Table 1. We will drag them on to the designer from the left hand side palette.

1. Drag and drop the components mentioned in Table 1 to the viewer.
2. Visible components will be seen by you while the non-visible components will be located beneath the viewer under the tag 'Non-visible'.
3. For the application's name, set the 'Screen1' title property to 'SlidingPuzzle'.
4. All images need to be put within the vertical arrangement so as to keep them aligned in a square view.
5. If you have dragged and placed everything, the layout will look something like what's shown in Figure 1.
6. Make the necessary property changes like we did in changing the text property for the label and button components.
7. Renaming the components helps to identify them in the block editor.
8. Your graphical user interface is ready. The application will look exactly like what's shown in Figure 2, after installation in the device.
9. The hierarchy of the components dragged to the designer is shown in Figure 3.

If you are confused by the designer and the components viewer, let me discuss it a bit more. Here is the hierarchy that we have placed for our application.

Table 1

Component name	Purpose	Location
1 Label	To display a label	Palette-->User Interface-->Label
2 Button	To trigger events	Palette-->User Interface-->Button
3 Horizontal arrangement	To arrange the child components	Palette-->Layout-->Horizontal Arrangement
4 Table arrangement	To arrange the child components	Palette-->Layout-->Table Arrangement
5 Clock	To have time control of events	Palette-->Sensors-->Clock
6 Notifier	To display on-screen information	Palette-->User Interface-->Notifier
7 Canvas	To enable drawing	Palette--> Drawing and Animation--> Canvas
8 Image	To display images	Palette-->User Interface-->Image
9 TinyDB	To store data	Palette-->Storage-->TinyDB